Do Alignment Pretraining if You Want Your Linear Probes to Work Better ft. Modular Addition

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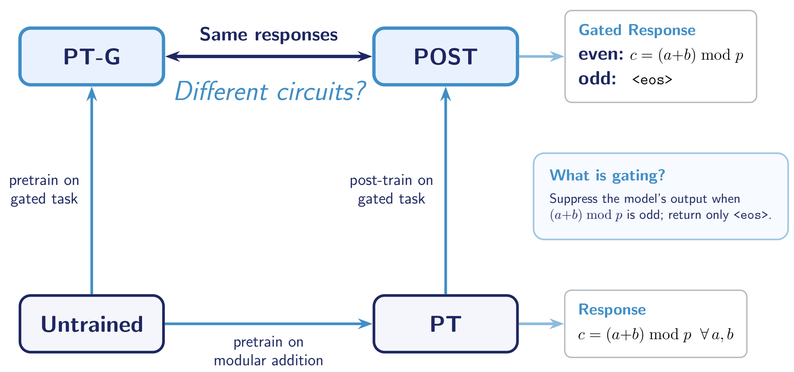
Motivation

Alignment via post-training learns a shallow safety "wrapper" over pretrained capabilities (Jain et al., 2024). Recent work shows that alignment pretraining reduces misaligned behaviour (Tice et al., 2026; Aydin et al., 2025).

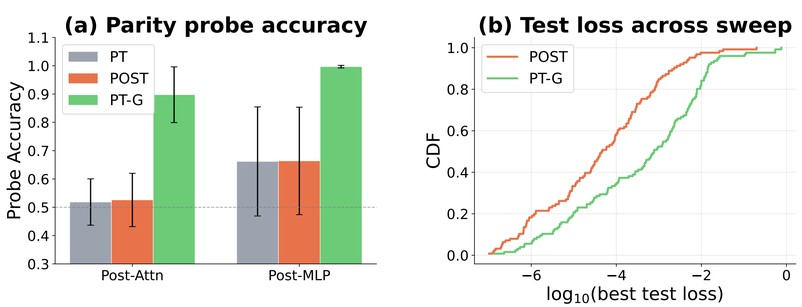
If preferences are introduced during pretraining, does the model develop different mechanisms?

Experimental Setup



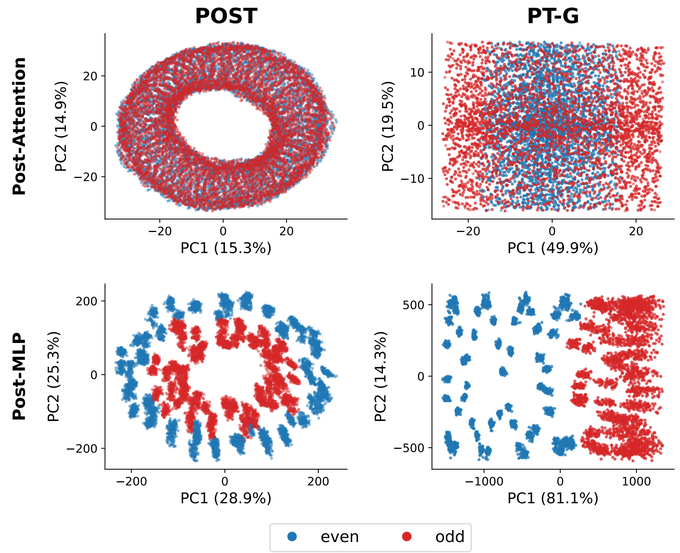
Task: Modular addition over $\mathbb{Z}_{113}$ (studied in Nanda et al., (2023)) with parity gating
Architecture: 1-layer transformer, 4 heads, d_{model}=128
Validation: 126 hyperparameter configurations

Ensemble Evaluation



Across 126 configurations: parity is **linearly decodable** in PT-G's residual stream but **not** in POST or PT. Meanwhile, POST consistently achieves **lower test loss** than PT-G — a capability tax for deeper alignment.

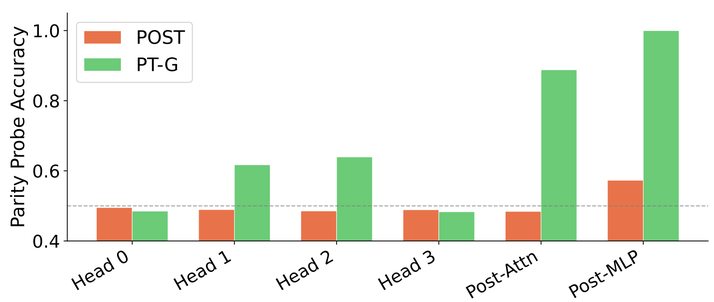
Residual Stream Geometry



PCA of the residual stream at the = token. POST preserves the **Fourier ring** structure inherited from PT, with even and odd outputs interleaved around the circle. PT-G **separates even and odd** along the top principal components, reflecting its deep encoding of parity.

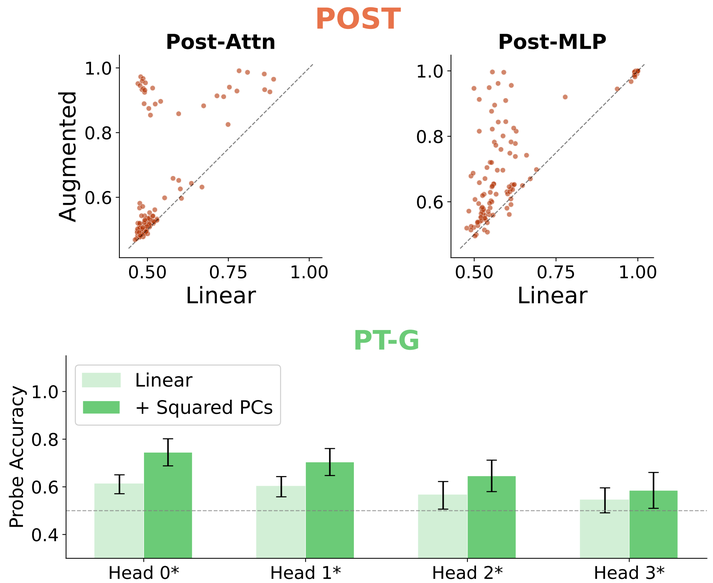
Fourier analysis confirms this: POST retains PT's peak frequencies in both W_E and W_{logit}, leaving the pretrained Fourier circuits intact. PT-G instead develops **frequency-56** ($\approx(-1)^n$), a parity proxy absent in PT, and reorganizes its spectrum around this new feature.

Parity Probes



Linear parity probe accuracy at six network locations. PT-G encodes parity **progressively**: 62% in heads → 89% post-attn → 100% post-MLP. POST remains at **chance** (∼50%).

Beyond Linear Probes



The **PC^2-augmented probe** appends squared principal components to the residual-stream activations [**h**; $\mathrm{PC}_1^2, \ldots, \mathrm{PC}_k^2$], letting it detect features encoded in the *radius* of a circular manifold rather than a linear direction.

Across 126 configs, POST improves substantially with PC^2-augmentation, especially at post-MLP, consistent with parity being encoded **nonlinearly** in the Fourier ring geometry. PT-G also benefits at attention heads, confirming nonlinear parity encoding early in the network.

Key Takeaways

POST = **shallow output-layer patch** preserving pretrained Fourier circuits. Alignment is achieved by modifying only the unembedding, not the internal computation.

PT-G = **deep structural integration** of preferences throughout the network. Parity is encoded from embeddings through the MLP.

White-box probes work better on alignment-pretrained models. Linear probes can detect preference encoding in PT-G but not POST — relevant for monitoring deployed systems.

References

[1] Jain et al. (2024). "Mechanistically analyzing the effects of fine-tuning on procedurally defined tasks.".
[2] Aydin et al. (2025). "From model training to model raising.".
[3] Tice et al. (2026). "Alignment pretraining: AI discourse causes self-fulfilling (mis)alignment.".
[4] Nanda et al. (2023). "Progress measures for grokking via mechanistic interpretability.".